

Dr. Rachele Ossola

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Education

2016 – 2021	Ph.D. in Environmental Chemistry (Prof. K. McNeill), ETH Zurich (Switzerland)
2014 – 2016	Master in Biological and Molecular Chemistry (5.92/6.00), EPFL (Switzerland)
2011 – 2014	Bachelor in Chemistry (110 <i>and honors</i> /110), University of Milan (Italy)

Research Experience

02/2022 – present	SNSF Postdoctoral Fellow in the Atmospheric Chemistry and Biogeochemistry group (Prof. D. Farmer), Colorado State University (Colorado, USA)
11/2016–11/2021	Ph.D. Candidate and Research Assistant in the Environmental Chemistry Group (Prof. K. McNeill), ETH Zurich (Switzerland)
09 – 12/2015	Student Assistant in the Environmental Chemistry Group (Prof. T. Kohn), EPFL (Switzerland)
07 – 09/2015	Student Assistant in the Environmental Chemistry Group (Prof. K. McNeill), ETH Zurich (Switzerland)

Main awards

2022	ETH Medal for doctoral thesis Awarded by ETH Zurich for “her outstanding contribution to our understanding of aquatic photochemical processes and the marine sulfur cycle”
2020	“C. Ellen Gonter Environmental Chemistry Award” Highest award given to students by the division of Environmental Chemistry of the American Chemical Society
2016	“EPFL Master” prize Second-best grade average in the EPFL Master (all sections considered)
2016	“Syngenta Montey (Chimie)” prize Best grade average in Molecular and Biological Chemistry in the section of Chemistry and Chemical engineering for the master cycle + master project
2016	“BASF Montey SA (Chimie)” prize Best master project in Molecular and Biological Chemistry

Main grants and fellowships

2023	EcoCore mini-grant (7'753 USD)
2022 – 2024	Swiss National Science Foundation Postdoc Mobility Fellowship (112'000 CHF) + Gender Equality Grant (2'000 CHF)

2011 – 2014 Italian Chemical Society National Scholarship (12'000 €)

Professional service

- 2023 – present Member of the Environmental Effects Assessment Panel (EEAP) of the United Nations Environmental Program (UNEP)
- 2021 – present Co-author for the United Nations Environmental Program (UNEP) Environmental Effects Assessment Panel (EEAP)
2023 – Working group 3 (Terrestrial Ecosystems and Biogeochemical Cycles); Participation at the UNEP EEAP meeting in Fort Collins, Colorado (Sept 17 – 24, 2023)
2022 – Working Groups 3 (Terrestrial Ecosystems) and 4 (Aquatic Ecosystems); Participation at the UNEP EEAP meeting in Kalmar, Sweden (Sept 18 – 27, 2022); Preparation of the Q&A document
2021 – Working Group 5 (Biogeochemistry)
- 2020 – present Reviewer for *Environmental Science and Technology*, *Environmental Science and Technology Letters*, *Science of the Total Environment*, *Atmospheric Chemistry and Physics*, and *Limnology and Oceanography Letters*

Teaching and mentoring

- 2018 – 2021 Teaching Assistant in "Practical Training in Biogeochemistry" (701-0420-01L), ETH Zurich
- 2018 – 2020 Teaching Assistant in "Organic Chemistry" (701-0225-00L), ETH Zurich
- 2017 – 2021 Teaching Assistant in "Laboratory Course: Elementary Chemical Techniques" (529-0030-00L), ETH Zurich
- 2017 – 2022 Student mentor
Baptiste Clerc (09/2017 – 02/2019), Richard Gruseck (09/2018 – 09/2019), Saskia Läubli (03 – 09/2019), Elvira Rudin (09/2020 – 09/2021), Audrey Lawrence (07 – 08/2022), Rose Rossell (12/2022 – now), Cameron Osburn (03/2023 – now)

Publications [\[google scholar profile\]](#)

Scientific articles

14. P. J. Neale, C. E. Williamson, A. T. Banaszak, D.-P. Häder, S. Hylander, **R. Ossola**, K. Rose, S.-Å. Wängberg, R. Zepp, "The response of aquatic ecosystems to the interactive effects of stratospheric ozone depletion, UV radiation, and climate change", *Photochemical & Photobiological Sciences* 22, 1093–1127 (2023)
13. P. W. Barnes, T. M. Robson, R. G. Zepp, J. F. Bornman, M. A. K. Jansen, **R. Ossola**, Q.-W. Wang, S. A. Robinson, B. Foereid, A. R. Klekociuk, J. Martinez-Abaigar, W.-C. Hou, N. D. Paul, "Interactive effects of changes in UV radiation and climate on terrestrial ecosystems, biogeochemical cycles, and feedbacks to the climate system", *Photochemical & Photobiological Sciences* 22, 1041–1091 (2023)
12. **R. Ossola***, R. Gruseck, J. Houska, A. Manfrin, M. Vallieres, K. McNeill*, "Photochemical production of carbon monoxide from dissolved organic matter: The role of lignin methoxyarene functional groups", *Environmental Science & Technology* 56 (18), 13449–13460 (2022) [* co-corresponding authors]
11. J. P.W. Barnes, T.M. Robson, P.J. Neale, C.E. Williamson, R.G. Zepp, S. Madronich, S.R. Wilson A.L. Andradý, A.M. Heikkilä, G.H. Bernhard, A.F. Bais, R.E. Neale, J.F. Bornman, M.A.K. Jansen, A.R. Klekociuk, J. Martinez-

Abaigar, S.A. Robinson, Q.-W. Wang, A.T. Banaszak, D.-P. Häder, Samuel Hylander, K.C. Rose, S.-Å. Wängberg, B. Foereid, W.-C. Hou, **R. Ossola**, N.D. Paul, J.E. Ukpebor, M.P.S. Andersen, J. Longstreth, T. Schikowski, K.R. Solomon, B. Sulzberger, L.S. Bruckman, K.K. Pandey, C.C. White, L. Zhu, M. Zhu, P.J. Aucamp, J.B. Liley, R.L. McKenzie, M. Berwick, S.N. Byrne, L.M. Hollestein, R. M. Lucas, C.M. Olsen, L.E. Rhodes, S. Yazar, A.R. Young, "Environmental effects of stratospheric ozone depletion, UV radiation, and interactions with climate change: UNEP Environmental Effects Assessment Panel, Update 2021", *Photochemical & Photobiological Sciences* 21, 275-301 (2022)

10. J. M. Buth, **R. Ossola**, S. Partanen, K. McNeill, W.A. Arnold, M. O'Connor, D. E. Latch, "Kinetics and pathways of the aqueous photolysis of pharmaceutical pollutants: A versatile laboratory or remote learning investigation", *Journal of Chemical Education* 98 (7), 2411-2418 (2021)

9. **R. Ossola**, O. Jönsson, K. J. Moor, K. McNeill, "Singlet oxygen quantum yields in environmental waters", *Chemical Reviews* 121 (7), 4100-4146 (2021)

8. **R. Ossola**, B. Clerc, K. McNeill, "Mechanistic insights into dissolved organic sulfur photomineralization through the study of cysteine sulfinic acid", *Environmental Science & Technology* 54 (20), 13066-13076 (2020)

7. **R. Ossola**, J. Tolu, B. Clerc, P. R. Erickson, L. H. E. Winkel, K. McNeill, "Photochemical formation of sulfate and methanesulfonic acid from dissolved organic sulfur", *Environmental Science & Technology* 53 (22), 13191-13200 (2019)

6. N. Borduas-Dedekind, **R. Ossola**, R. O. David, L. Boynton, V. Weichlinger, K. McNeill, Z. A. Kanji, "Photomineralization mechanism changes the ability of dissolved organic matter to activate cloud droplets and to nucleate ice crystals", *Atmospheric Chemistry and Physics* 19, 12397-12412 (2019)

5. **R. Ossola**, M. Schmidt, P. R. Erickson, K. McNeill, "Furan carboxamides as model compounds to study the competition between two modes of indirect photochemistry", *Environmental Science & Technology* 53 (16), 9594-9603 (2019)

4. A. Colombo, **R. Ossola**, M. Magni, D. Roberto, D. Jacquemin, C. Castellano, F. Demartin, C. Dragonetti, "Intriguing C-H ... Cu interactions in bis-(phenantroline) Cu(I) redox mediators for dye-sensitized solar cells", *Dalton Transactions* 47, 1018-1022 (2018)

3. Q. Zhong, A. Carratalà, **R. Ossola**, V. Bachmann, T. Kohn, "Cross-resistance of UV- or chlorine dioxide-resistant Echovirus 11 to other disinfectants", *Frontiers in Microbiology* 8, 1928 (2017)

2. M. E. F. Bouduban, A. Burgos-Caminal, **R. Ossola**, J. Teuscher, J. E. Moser, "Energy and charge transfer cascade in methylammonium lead bromide perovskite nanoparticle aggregates", *Chemical Sciences* 8 (6), 4371-4380 (2017)

1. E. Appiani*, **R. Ossola***, D. E. Latch, P. R. Erickson, K. McNeill, "Aqueous singlet oxygen reaction kinetics of furfuryl alcohol: effect of temperature, pH, and salt content", *Environmental Science: Processes & Impacts* 19 (4), 507-516 (2017) [*shared first authorship]

Assessments and book chapters

4. C. Faiola, **R. Ossola***, D. F. McGlynn, "The role of BVOCs and plant surfaces in the formation and scavenging of ozone and particulate matter, including secondary organic aerosol", in "Biogenic volatile organic compounds and climate change", editors F. Brilli and S. Dicesare [* corresponding author], *in press*

3. United Nations Environment Programme (UNEP), "Questions and Answers about the Effects of Ozone Depletion, UV Radiation, and Climate on Humans and the Environment. Supplement of the 2022

Assessment Report of the UNEP Environmental Effects Assessment Panel”, 25 pp.; UNEP: Nairobi (2023) [[here](#); role: Production Team Co-coordinator]

2. United Nations Environment Programme (UNEP), “Environmental Effects of Stratospheric Ozone Depletion, UV Radiation, and Interactions with Climate Change. 2022 Assessment Report”, 375 pp.; UNEP: Nairobi (2023) [[here](#); role: co-author for Chapters 4 and 5]

1. United Nations Environment Programme (UNEP), “Environmental Effects of Stratospheric Ozone Depletion, UV Radiation, and Interactions with Climate Change: UNEP Environmental Effects Assessment Panel, Update 2021”, 17 pp.; UNEP: Nairobi (2023) [[here](#); role: co-author]

Selected presentations at scientific conferences and invited seminars

R. Ossola, “Investigating the chemistry of complex organic mixtures: Carbon monoxide production from the photodegradation of dissolved organic matter”, invited seminar at Carnegie Mellon University, April 29, 2022 (virtual)

R. Ossola, O. M. Jönsson, K. Moor, K. McNeill, “Critical evaluation of measurement approaches for apparent singlet oxygen quantum yields”, ACS Spring 2021 Online National Meeting and Exposition, April 5 – 30, 2021

R. Ossola, R. Gruesek, A. Manfrin, K. McNeill, “Photochemical production of carbon monoxide from dissolved organic matter: the role of lignin methoxyarene functional groups”, ACS Fall 2020 Online National Meeting and Exposition, August 17 – 20, 2020 (invited talk)

R. Ossola, J. Tolu, B. Clerc, P. R. Erickson and K. McNeill, “Photochemical production of sulfate and methanesulfonic acid from dissolved organic sulfur”, ACS Spring 2019 National Meeting and Exposition, Orlando, Florida, March 31 – April 4, 2019

R. Ossola, B. Clerc and K. McNeill, “Understanding sulfate production from photosensitized cysteine degradation”, ACS Spring 2019 National Meeting and Exposition, Orlando, Florida, March 31 – April 4, 2019

Main outreach activities

2022 – present	Production of outreach material for the United Nation Environmental Program
2020	Invited speaker at the outreach event “Settimana della scienza, I Edizione”, Domodossola (Italy), January 20 – 24, 2020
2019 – present	Italian-English translator for the International Younger Chemistry Network
2018 – 2019	Blog writer for “Environmental Science: Water Research & Technology” and “Environmental Science: Processes & Impacts”
2018	Volunteer for the “Art&Science Collaborative”
2017	Organization of the McNeill group panel for “Scientifica” (biennial outreach event organized by ETH Zürich), Zurich (CH), September 1 – 3, 2017
2009 – 2012	Volunteer at the Association “G. V. Schiaparelli”, Varese (Italy)